cadence®

Certus Stylus Text Command Reference Manual

Product Version 26.10
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Contents

About This Manual

The Cadence® Stylus Implementation System family of products provides an integrated solution for an RTL-to-GDSII design flow. This manual describes the syntax for the Stylus Implementation System (Stylus) text commands.

Audience

This manual is written for experienced designers of digital integrated circuits. Such designers must be familiar with design planning, placement and routing, block implementation, chip assembly, and design verification. Designers must also have a solid understanding of UNIX and Tcl/Tk programming.

Conventions Used in This Manual

This section describes the typographic and syntax conventions used in this manual.

`text`	Indicates text that you must type exactly as shown. For example: `analyze_connectivity -analyze all`
`text`	Indicates information for which you must substitute a name or value. In the following example, you must substitute the name of a specific file for *`configfile`*: `wroute filename` *`configfile`*
text	Indicates the following: • Text found in the graphical user interface (GUI), including form names, button labels, and field names • Terms that are new to the manual, are the subject of discussion, or need special emphasis • Titles of manuals

`[ ]`	Indicates optional arguments. In the following example, you can specify none, one, or both of the bracketed arguments: `command [-arg1] [arg2 value]`
`[ | ]`	Indicates an optional choice from a mutually exclusive list. In the following example, you can specify any of the arguments or none of the arguments, but you cannot specify more than one: `command [arg1 | arg2 | arg3 | arg4]`
`{ | }`	Indicates a required choice from a mutually exclusive list. In the following example, you must specify one, and only one, of the arguments: `command {arg1 | arg2 | arg3}`
`{[ ] [ ]}`	Indicates a required choice of one or more items in a list. In the following example, you must choose one argument from the list, but you can choose more than one: `command {[arg1] [arg2] [arg3]}`
`{ }`	Indicates curly braces that must be entered with the command syntax. In the following example, you must type the curly braces: `command arg1 {x y}`
`...`	Indicates that you can repeat the previous argument.
. . .	Indicates an omission in an example of computer output or input.
Command - Subcommand	Indicates a command sequence, which shows the order in which you choose commands and subcommands from the GUI menu. In the following example, you choose *Power* from the menu, then *Power Planning* from the submenu, and then *Add Rings* from the displayed list: *Power - Power Planning - Add Rings* This sequence opens the *Add Rings* form.

Related Documents

For more information about Certus, see the following documents. You can access these and other Cadence documents using Doc Assistant (online documentation system):

- *New Features in Certus Stylus*
  Provides information about new features in the release of Certus.
- *Certus Stylus User Guide*
  Provides information about using various Certus features.
- README file
  Contains installation, compatibility, and other prerequisite information for this release. You can read this file online at downloads.cadence.com.

For a complete list of documents provided with this release, see the Cadence Help online documentation system.

1

Certus ECO Commands and Attributes

The list of Certus ECO commands and attributes that can be specified at the Certus command prompt is provided below:

distribute_certus_command

```
distribute_certus_command
[-help]
<command> [-designs <string>]
```

Queries the physical attributes or properties from Certus clients or verifies the placement status of a few instances in the design. The `distribute_certus_command` command can be run at any point after `start_distributed_eco_clients` is specified. The tool waits for the design to be loaded on the Certus clients before running the commands that are passed through `distribute_certus_command`.

Note:

The commands passed to `distribute_certus_command` should not impact routing, change the don't_touch/don't_use attribute, or trigger a new parasitic extraction. In addition, the tool should not reset any parasitics or generate any timing reports. These are not included as it may desynchronize the Certus client manager from the Certus clients.

Parameters

`-help`	Outputs a brief description that includes type and default information for each `distribute_certus_command` parameter. For a detailed description of the command and all its parameters, use the man command: `man distribute_certus_command`.
`<command>`	Specifies the command for which you want to query the physical attributes/properties from Certus clients.
`-designs <string>`	Specifies the list of partitions for which `distribute_certus_command` is executed.

Examples

- The following command queries `report_annotated_parasitics` from each partition and sends the output to the Certus clients. The output shows a summary report, which is a list of annotated and unannotated nets:

  ```
  certus> distribute_certus_command {report_annotated_parasitics}
  ```

  The following output is generated:

  ```
  *info: output from client dtmf_recvr_core(0).

  ===============================================


  ####################################################################

  # report_annotated_parasitics: Tue Feb 25 01:09:41 2025

  #   view: setup_func

  #   -list_not_annotated

  ####################################################################
  ```

```
#

# list syntax: type: net_name [nrCap Cap nrXCap XCap nrRes Res]

# unit: pF, Ohm


# Summary of Annotated Parasitics:

+--------------------------------------------------------------------------------+

|    Net Type          |    Count   |   Annotated (%)     | Not Annotated (%)   |

+---------------------+------------+---------------------+---------------------+

| total                |       1010 |        881 87.23%   |        129 12.77%   |

+---------------------+------------+---------------------+---------------------+

| assign (*)           |          3 |          0  0.00%   |          3 100.00%   |

| 0-term:floating (*) |         23 |          0  0.00%   |         23 100.00%   |

| no drive (*)         |         33 |          0  0.00%   |         33 100.00%   |

| 1-term:no load       |         57 |          0  0.00%   |         57 100.00%   |

| real net (complete) |        861 |        848 98.49%   |         13  1.51%   |

| real net (broken)    |         33 |         33 100.00%   |          0  0.00%   |

| zero capacitance net|          0 |          0  0.00%   |          0  0.00%   |

+--------------------------------------------------------------------------------+

Note for Net Types marked "(*)":  Such nets are never timed, but reported here for
informational purpose.

+------------------------------------------------------------------+

| Annotated |    Res (KOhm)    |    Cap (pF)     |    XCap (pF)    |

+-----------+------------------+-----------------+-----------------+

| Count     |         10514    |         10782   |          2480   |

| Value     |       59.9570    |        4.1717   |        0.8583   |

+------------------------------------------------------------------+
```

```
*info: output from client ptn_wrapper(1).

===============================================


#####################################################################

# report_annotated_parasitics: Tue Feb 25 01:09:41 2025

#   view: ptn_wrapper_1_setup_func

#   -list_not_annotated

#####################################################################

#

# list syntax: type: net_name [nrCap Cap nrXCap XCap nrRes Res]

# unit: pF, Ohm


# Summary of Annotated Parasitics:

+--------------------------------------------------------------------------------+

|    Net Type          |    Count   |   Annotated (%)     | Not Annotated (%)   |

+----------------------+------------+---------------------+---------------------+

| total                |        241 |        139 57.68%   |        102 42.32%   |

+----------------------+------------+---------------------+---------------------+

| 0-term:floating (*)  |         37 |          0  0.00%   |         37 100.00%   |

| no drive (*)         |         32 |          0  0.00%   |         32 100.00%   |

| 1-term:no load       |         33 |          0  0.00%   |         33 100.00%   |

| real net (complete)  |        139 |        139 100.00%   |          0  0.00%   |

| real net (broken)    |          0 |          0  0.00%   |          0  0.00%   |

| zero capacitance net|          0 |          0  0.00%   |          0  0.00%   |

+--------------------------------------------------------------------------------+

Note for Net Types marked "(*)":  Such nets are never timed, but reported here for
informational purpose.
```

```
+-------------------------------------------------------------------+
| Annotated |    Res (KOhm)    |    Cap (pF)     |    XCap (pF)    |
+-----------+------------------+-----------------+-----------------+
| Count     |          1592    |          1658   |          272    |
| Value     |        8.0015    |        0.5732   |       0.0827    |
+-------------------------------------------------------------------+

*info: output from client tdsp_core(2).
===============================================

####################################################################
# report_annotated_parasitics: Tue Feb 25 01:09:41 2025
#   view: setup_func
#   -list_not_annotated
####################################################################
#
# list syntax: type: net_name [nrCap Cap nrXCap XCap nrRes Res]
# unit: pF, Ohm

# Summary of Annotated Parasitics:
+--------------------------------------------------------------------------------+
|    Net Type         |    Count    |   Annotated (%)      | Not Annotated (%)   |
+---------------------+-------------+----------------------+---------------------+
| total               |        3547 |       2553 71.98%    |        994 28.02%   |
+---------------------+-------------+----------------------+---------------------+
| assign (*)          |          59 |          0  0.00%    |         59 100.00%   |
| 0-term:floating (*) |         876 |          0  0.00%    |        876 100.00%   |
| no drive (*)        |          56 |          0  0.00%    |         56 100.00%   |
```

```
| 1-term:no load      |          3 |          0  0.00%   |          3 100.00%   |

| real net (complete) |       2553 |       2553 100.00%   |           0  0.00%   |

| real net (broken)   |          0 |          0  0.00%   |          0  0.00%   |

| zero capacitance net|          0 |          0  0.00%   |          0  0.00%   |

+--------------------------------------------------------------------------------+

Note for Net Types marked "(*)":  Such nets are never timed, but reported here for
informational purpose.


+------------------------------------------------------------------+

| Annotated |    Res (KOhm)    |    Cap (pF)     |    XCap (pF)    |

+-----------+-----------------+-----------------+-----------------+

| Count     |         38329   |         38418   |          5288   |

| Value     |      145.6864   |        5.9198   |        0.7150   |

+------------------------------------------------------------------+
```

- The following command queries `report_annotated_parasitics` from the selected partitions `{ptn_wrapper}`:

```
certus> distribute_certus_command {report_annotated_parasitics} -designs
{ptn_wrapper}
```

The following output is generated:

```
*info: output from client ptn_wrapper(1).

===============================================


####################################################################

# report_annotated_parasitics: Tue Feb 25 01:14:42 2025

#   view: ptn_wrapper_1_setup_func

#   -list_not_annotated

####################################################################
```

```
#
# list syntax: type: net_name [nrCap Cap nrXCap XCap nrRes Res]
# unit: pF, Ohm

# Summary of Annotated Parasitics:
+----------------------------------------------------------------------------------+
|    Net Type          |    Count   |   Annotated (%)      | Not Annotated (%)    |
+---------------------+------------+---------------------+---------------------+
| total               |        241 |        139 57.68%   |        102 42.32%   |
+---------------------+------------+---------------------+---------------------+
| 0-term:floating (*) |         37 |          0  0.00%   |         37 100.00%   |
| no drive (*)        |         32 |          0  0.00%   |         32 100.00%   |
| 1-term:no load      |         33 |          0  0.00%   |         33 100.00%   |
| real net (complete) |        139 |        139 100.00%   |          0  0.00%   |
| real net (broken)   |          0 |          0  0.00%   |          0  0.00%   |
| zero capacitance net|          0 |          0  0.00%   |          0  0.00%   |
+----------------------------------------------------------------------------------+
Note for Net Types marked "(*)":  Such nets are never timed, but reported here for
informational purpose.

+---------------------------------------------------------------+
| Annotated |    Res (KOhm)   |    Cap (pF)     |    XCap (pF)    |
+-----------+-----------------+-----------------+-----------------+
| Count     |          1592   |          1658   |           272   |
| Value     |        8.0015   |        0.5732   |        0.0827   |
+---------------------------------------------------------------+
```

- The following command queries `check_place` from each partition:

```
certus> distribute_certus_command {check_place}
```

The following output is generated:

```
*info: output from client dtmf_recvr_core(0).
===============================================

**WARN: (IMPSP-362):  Site 'CoreSite' has one std.Cell height, so ignoring its X-
symmetry.
Type 'man IMPSP-362' for more detail.
**Info: (IMPSP-307): Design contains fractional 17 cells.

Begin checking placement ... (start mem=2306.8M, init mem=2322.9M)
Region/Fence Violation:    1
*info: Placed = 766            (Fixed = 119)
*info: Unplaced = 0
Placement Density:1.36%(2262/165793)
Placement Density (including fixed std cells):1.50%(2500/166031)
Finished check_place (total: cpu=0:00:00.3, real=0:00:00.0; vio checks:
cpu=0:00:00.1, real=0:00:00.0; mem=2315.6M)

*info: output from client ptn_wrapper(1).
===============================================

**WARN: (IMPSP-362):  Site 'CoreSite' has one std.Cell height, so ignoring its X-
symmetry.
Type 'man IMPSP-362' for more detail.
**Info: (IMPSP-307): Design contains fractional 17 cells.

Begin checking placement ... (start mem=2345.8M, init mem=2359.7M)
*info: Placed = 115            (Fixed = 2)
*info: Unplaced = 0
Placement Density:0.81%(202/24977)
Placement Density (including fixed std cells):0.81%(202/24977)
Finished check_place (total: cpu=0:00:00.2, real=0:00:00.0; vio checks:
cpu=0:00:00.0, real=0:00:00.0; mem=2352.7M)

*info: output from client tdsp_core(2).
===============================================

**WARN: (IMPSP-362):  Site 'CoreSite' has one std.Cell height, so ignoring its X-
symmetry.
Type 'man IMPSP-362' for more detail.
**Info: (IMPSP-307): Design contains fractional 17 cells.

Begin checking placement ... (start mem=2396.5M, init mem=2423.3M)
*info: Placed = 2495
*info: Unplaced = 0
Placement Density:18.07%(7851/43437)
```

```
Placement Density (including fixed std cells):18.07%(7851/43437)
Finished check_place (total: cpu=0:00:00.3, real=0:00:00.0; vio checks:
cpu=0:00:00.1, real=0:00:00.0; mem=2410.2M)
```

Related Commands

- start_distributed_eco_clients

export_distributed_eco

```
export_distributed_eco
[-help]
[-designs <string>]
[-export_only innovus_dbs | sta_dbs] -out_dir <string>
[-overwrite]
```

Exports Innovus DBs and Tempus STA sessions in `out_dir` at the end of the Certus-based optimization.

Parameters

`-help`	Outputs a brief description that includes type and default information for each `export_distributed_eco` parameter.
`-designs <string>`	Specifies the list of modules for which an Innovus partition DB needs to be exported. *Default*: `all`
`-export_only innovus_dbs | sta_dbs`	Exports client Tempus sessions (`sta_dbs`) or Innovus DBs (`innovus_dbs`).
`-out_dir <string>`	Specifies the name of the output directory in which all data will be written. This output directory includes Innovus_DBs and STA_dbs directories.
`-overwrite`	Overwrites the existing `out_dir` if the same directory is provided again.

Examples

The following command saves Innovus DBs for all partitions in the output directory. These saved Innovus DBs can be used for the block-level flows to clean up any remaining routing DRCs or to generate Verilog/DEF. However, these saved Innovus DBS should not be used to generate any timing reports or to perform further timing/power optimization, as timing cannot be checked further at block-level. This is because Certus has made netlist updates based on a timing picture from the sub-system/full-chip level:

```
certus> export_distributed_eco -export_only {innovus_dbs} -out_dir output
```

Related Commands

- time_design_signoff

generate_summary

```
generate_summary
[-help]
[-logvfilename <string>]
[-outputfile <string>]
```

Displays the Certus log summary report in a text file. The log summary report includes the following details:

- Runtime details of the full flow.
- Tool versions for Certus Manager, Certus Clients, and Tempus Clients.
- Active views for each Tempus Client.
- Partition Name, Netlist Size, the Number of Replications, and Log Pointer for each Certus Client.
- Number of CPUs enabled for Certus Manager, Certus Clients, and Tempus Clients.
- Size of the netlist on the Certus Manager, Certus Clients, and Tempus Clients.
- RC annotation summary for each partition loaded as a Certus client.
- QoR/TAT/Memory summary from `time_design_signoff`, ECO optimization, and `implement_distributed_eco` steps.
- Mean displacement, Max displacement, Instances moved, and Errors in `place_detail` for each partition.
- Final Routing DRC after each `implement_distributed_eco` call for each partition.
- Initial density (from the first optimization step) and the final density (from the last optimization step) for each partition.
- List of error codes and DIAGs for Certus Manager, Tempus Clients, and Certus Clients.

Parameters

`-help`	Outputs a brief description that includes type and default information for each `generate_summary` parameter. For a detailed description of the command and all of its parameters, use the `man` command: `man generate_summary.`
`-logvfileName <string>`	Specifies the name of the log file. By default, the log file name is `certus.logv`. **Note:** This parameter is optional.
`-outputfile <string>`	Specifies the name of the output file, which includes the log summary details for the specified log file. For the log file, `certus.logv`, the default log summary file name is `certus.logv.summary.txt.` **Note:** This parameter is optional.

Examples

- The following command generates the summary report, `certus.logv.summary.txt,` for the Certus log file, `certus.logv`:

  ```
  certus> generate_summary
  ```

- The sample script below performs distributed hold optimization in full-flat mode and ends with the ECO implementation step. This step does ECO routing and parasitic extraction on each Certus client, followed by a timing update with updated parasitics on the timing clients.

  ```
  set_db eco_setup_view_list "V1 V2"
  set_db eco_hold_view_list "V1 V2"
  set_db eco_dynamic_view V1
  set_db eco_leakage_view V2
  set_multi_cpu_usage -local_cpu 12 -remote_host 2 -cpu_per_remote_host 8
  set_db opt_signoff_distributed_optimization_block_restore_file blkRestore.txt
  distribute_read_db -restore_design {save_db_dmmmc_V1 save_db_dmmmc_V2} -out_dir
  ```

```
paradime_Output
set_multi_cpu_usage -local_cpu 12 -remote_host 3 -cpu_per_remote_host 8
start_distributed_eco_clients
set_db opt_signoff_verbose true
time_design_signoff -report_dir INITIAL_TDS -report_power
set_db opt_signoff_eco_file_prefix ECO/drv
opt_signoff -drv
set_db opt_signoff_eco_file_prefix ECO/setup
opt_signoff -setup
set_db opt_signoff_eco_file_prefix ECO/hold
opt_signoff -hold
implement_distributed_eco
time_design_signoff -report_dir FINAL_TDS -report_power
generate_summary
```

The following is a sample of the resulting text report `(certus.logv.summary.txt):`

```
######################################################
#
#  Cadence Certus(TM) Solution Summary File
#  Created on Wednesday, Jul 02, 2025 02:07:33
#
#######################################################
Run Started on:    Wed, Jul 02, 2025 01:56:21
Run Terminated on: Wed, Jul 02, 2025 02:04:23
Full Flow Runtime: 00:08:02


+-------------------------+--------------+
|        Process          |   Version    |
+-------------------------+--------------+
| Certus Manager Version | 25.11-e070_1 |
| Certus Client Version  | 25.11-e070_1 |
| Tempus Client Version  | 25.11-e070_1 |
+-------------------------+--------------+


+----------------+--------------------------------------------------------------+-
----------------------------------------------------------------------------+
|Tempus Client(s)|                            View(s)                             |
                                Log Pointer                                |
+----------------+--------------------------------------------------------------+-
----------------------------------------------------------------------------+
```

```
| tempus_client0 |                          V1                              |
    paradime_Output/1872625_668c8f94-2bf4-4821-8238-2f3c26a7ed9a_0.log      |
| tempus_client1 |                          V2                              |
    paradime_Output/1872625_668c8f94-2bf4-4821-8238-2f3c26a7ed9a_1.log      |
+----------------+--------------------------------------------------------------+-
----------------------------------------------------------------------------+



+-----------------+-----------------+-------------+--------------------+--------
----------------------------------+
|Certus Client(s) | Partition Name  | Netlist Size | Replicated Modules  |
      Log Pointer               |
+-----------------+-----------------+-------------+--------------------+--------
----------------------------------+
| Certus_client0  | dtmf_recvr_core |          766 |          1           |
 DistOpt/1872625_0.log          |
| Certus_client1  |   ptn_wrapper   |          113 |          2           |
 DistOpt/1872625_1.log          |
| Certus_client2  |    tdsp_core    |         2495 |          1           |
 DistOpt/1872625_2.log          |
+-----------------+-----------------+-------------+--------------------+--------
----------------------------------+


+------------------+-----------------+

|     Process      |  CPUs enabled   |

+------------------+-----------------+

|  certus_manager  |       12        |

|  tempus_client0  |        8        |

|  tempus_client1  |        8        |

|  certus_client0  |        8        |

|  certus_client1  |        8        |

|  certus_client2  |        8        |

+------------------+-----------------+


+------------------+----------+
|   Netlist Size   |  StdCell |
+------------------+----------+
```

```
|  Tempus Client   |   3490   |
|  Certus Manager  |   3490   |
+------------------+----------+


+-----------------+-----------+--------------------------+---------------------+--
-----+---------------+------------------+
| Block           | RC Corner | View                     | Net Type            |
Count | Annotated (%) | Not Annotated (%) |
+-----------------+-----------+--------------------------+---------------------+--
-----+---------------+------------------+
| dtmf_recvr_core | rcMax     | V1                       | real net (complete) |
861   | 848 98.49%    | 13 1.51%         |
| ptn_wrapper     | rcMax     | ptn_wrapper_1_V1         | real net (complete) |
139   | 139 100.00%   | 0 0.00%          |
| tdsp_core       | rcMax     | V1                       | real net (complete) |
2553  | 2553 100.00%  | 0 0.00%          |
+-----------------+-----------+--------------------------+---------------------+--
-----+---------------+------------------+



#####################################################################QoR
Summary#####################################################################
Legend:  H: Hold; S: Signal Nets; C: Clock Nets
time_design_signoff reports summary for all Path Group
+-----------------------------+------------+------------+------------+-----------
+------------+------------+------------+------------+------------+------------+---
---------+------------+------------+------------+------------+------------+-------
-----+------------+------------+-----------------------------+
|            Step             |    WNS     |    TNS     |     VP     |   WNS(H)
|   TNS(H)   |   VP(H)    |  m_cap(S)  |  m_cap(C)  | m_tran(S)  | m_tran(C)  |
LkgPwr    |   IntPwr   |   SwPwr    |    TPwr    |  Runtime   |   Memory   |
Added     |  Resized   |  Deleted   |            Step             |
+-----------------------------+------------+------------+------------+-----------
+------------+------------+------------+------------+------------+------------+---
---------+------------+------------+------------+------------+------------+-------
-----+------------+------------+-----------------------------+
|    time_design_signoff 1    |   -2.455   |  -66.882   |     49     |   -5.287
|  -189.695  |    186     |    0(0)    |    3(3)    |    1(8)    |   5(43)    |
0.0005    |   0.3393   |   0.3207   |   0.6605   | 0:00:13.0  |  2856.9M   |
     |            |            |    time_design_signoff 1    |
|  opt_signoff -drv Initial   |   -2.455   |  -66.882   |     49     |   -5.287
```

```
|  -189.695  |    186     |    0(0)    |    3(3)    |   4(22)    |   5(43)    |
         |            |            |            |            |            |
     |            |            |  opt_signoff -drv Initial   |
|   opt_signoff -drv Final    |   -2.455   |  -66.882   |     49     |   -5.287
|  -189.695  |    186     |    0(0)    |    3(3)    |    0(0)    |   5(43)    |
         |            |            |            | 0:00:52.0  |  4072.3M   |     3
     |     1      |            |   opt_signoff -drv Final    |
| opt_signoff -setup Initial  |   -2.455   |  -66.882   |     49     |   -5.287
|  -189.695  |    186     |    0(0)    |    3(3)    |    0(0)    |   5(43)    |
         |            |            |            |            |            |
     |            |            | opt_signoff -setup Initial  |
|  opt_signoff -setup Final   |   -2.426   |  -38.758   |     16     |   -5.287
|  -189.694  |    186     |    0(0)    |    3(3)    |    0(0)    |   5(43)    |
         |            |            |            | 0:00:09.0  |  4008.3M   |     3
     |     27     |            |  opt_signoff -setup Final   |
|  opt_signoff -hold Initial  |   -2.426   |  -38.758   |     16     |   -5.287
|  -189.694  |    186     |    0(0)    |    3(3)    |    0(0)    |   5(43)    |
         |            |            |            |            |            |
     |            |            |  opt_signoff -hold Initial  |
|   opt_signoff -hold Final   |   -2.426   |  -38.757   |     16     |   0.002
 |   0.000    |     0      |    0(0)    |    3(3)    |    0(0)    |   5(43)    |
         |            |            |            | 0:00:30.0  |  4054.3M   |    230
    |            |            |   opt_signoff -hold Final   |
|          ecoRoute           |            |            |            |
 |            |            |            |            |            |            |
         |            |            |            | 0:00:24.0  |            |
     |            |            |          ecoRoute           |
|       extractRC_time        |            |            |            |
 |            |            |            |            |            |            |
         |            |            |            | 0:00:13.0  |            |
     |            |            |       extractRC_time        |
|      tps_client_update      |            |            |            |
 |            |            |            |            |            |            |
         |            |            |            | 0:00:04.0  |            |
     |            |            |      tps_client_update      |
|    time_design_signoff 2    |   -2.282   |  -36.933   |     27     |   -0.254
|   -1.639   |     24     |    0(0)    |    3(3)    |    0(0)    |   4(41)    |
0.0006   |   0.448    |   0.3026   |   0.7512   | 0:00:10.0  |  3777.9M   |
     |            |            |    time_design_signoff 2    |
+-----------------------------+------------+------------+------------+------------
+------------+------------+------------+------------+------------+------------+---
---------+------------+------------+------------+------------+------------+-------
-----+------------+------------+-----------------------------+
```

```
################################################################### Time Design
Signoff Summary ################################################################
+-----------------------+-----------------------+---------------+---------------+-----
---------+---------------+---------------+---------------+
|         Step          |       Path Group      |      WNS      |      TNS      |
 VP      |    WNS(H)     |    TNS(H)     |     VP(H)     |
+-----------------------+-----------------------+---------------+---------------+-----
---------+---------------+---------------+---------------+
| time_design_signoff1 |          all          |    -2.455     |    -66.882    |
 49      |    -5.287     |   -189.695    |      186      |
| time_design_signoff1 |        reg2reg        |    -1.033     |    -27.880    |
 33      |    -4.098     |    -77.486    |      37       |
| time_design_signoff1 |         in2reg        |     1.744     |     0.000     |
  0      |    -5.287     |   -156.242    |      163      |
| time_design_signoff1 |        reg2out        |    -2.455     |    -39.002    |
 16      |     0.413     |     0.000     |       0       |
| time_design_signoff1 |         in2out        |     1.420     |     0.000     |
  0      |     0.594     |     0.000     |       0       |
+-----------------------+-----------------------+---------------+---------------+-----
---------+---------------+---------------+---------------+
| time_design_signoff2 |          all          |    -2.282     |    -36.933    |
 27      |    -0.254     |    -1.639     |      24       |
| time_design_signoff2 |        reg2reg        |    -0.090     |    -0.482     |
 11      |    -0.254     |    -1.026     |      12       |
| time_design_signoff2 |         in2reg        |     0.836     |     0.000     |
  0      |    -0.060     |    -0.612     |      12       |
| time_design_signoff2 |        reg2out        |    -2.282     |    -36.451    |
 16      |     0.416     |     0.000     |       0       |
| time_design_signoff2 |         in2out        |     0.306     |     0.000     |
  0      |     0.517     |     0.000     |       0       |
+-----------------------+-----------------------+---------------+---------------+-----
---------+---------------+---------------+---------------+


#################################################################### Drv Opt
####################################################################
DRV Optimization
+------------------------------------+----------------------+
|             Attribute              |        Value         |
+------------------------------------+----------------------+
```

```
|            Timing Mode              |       SOCV PBA      |
|           Analysis Mode             |        WFP SI       |
|               EOSDB                 |          On         |
| max_tran fixing rate nets (terms)   |    100.0%(100.0%)   |
| max_cap fixing rate nets (terms)    |      N/A%(N/A%)     |
+------------------------------------+---------------------+


+-------------------------+--------------------+-------------------+-----------------
+----------------------+
| Refine Place Movements | Mean displacement | Max displacement | Instances Moved
|       ERROR ID       |
+-------------------------+--------------------+-------------------+-----------------
+----------------------+
|     Certus_Client0      |      0.00 um       |      0.00 um      |       N/A
|         N/A          |
|     Certus_Client1      |        N/A         |       N/A         |       N/A
|         N/A          |
|     Certus_Client2      |      0.00 um       |      0.00 um      |       N/A
|         N/A          |
+-------------------------+--------------------+-------------------+-----------------
+----------------------+


Failure Reasons for Drv Opt:
+--------------------------------------------------+----------------+
|         Could not be fixed (Clock Nets )          | Number of nets |
+--------------------------------------------------+----------------+
|                     ClockNet                     |       5        |
+--------------------------------------------------+----------------+


################################################################### Setup Opt
###################################################################
Setup Optimization
+--------------------------+---------------------+
|        Attribute         |        Value        |
+--------------------------+---------------------+
|       Timing Mode        |       SOCV PBA      |
|      Analysis Mode       |        WFP SI       |
|          EOSDB           |          On         |
```

```
|     WNS fixing rate     |         1.2%         |
|     TNS fixing rate     |        42.1%         |
|     VP fixing rate      |        67.3%         |
+-------------------------+----------------------+


+------------------------+--------------------+-------------------+-----------------
+----------------------+
| Refine Place Movements | Mean displacement | Max displacement | Instances Moved
|       ERROR ID       |
+------------------------+--------------------+-------------------+-----------------
+----------------------+
|     Certus_Client0     |      0.00 um       |      0.00 um      |       N/A
|         N/A          |
|     Certus_Client1     |        N/A         |        N/A        |       N/A
|         N/A          |
|     Certus_Client2     |      0.00 um       |      0.00 um      |       N/A
|         N/A          |
+------------------------+--------------------+-------------------+-----------------
+----------------------+

Failure Reasons for Setup Opt:
+---------------------------------------------------+-----------------+
|         Could not be fixed ( Data Nets )          | Number of nets |
+---------------------------------------------------+-----------------+
|              SetupNotImprovingBuff                |       23        |
|              HoldWNSDegradationRsz                |        2        |
|               SetupNotImprovingDel                |        1        |
+---------------------------------------------------+-----------------+



################################################################## Hold Opt
##################################################################
Hold Optimization
+-------------------------+----------------------+
|        Attribute        |        Value         |
+-------------------------+----------------------+
|       Timing Mode       |       SOCV PBA       |
|      Analysis Mode      |        WFP SI        |
|          EOSDB          |         On           |
|     WNS fixing rate     |        100.0%        |
```

```
|     TNS fixing rate     |        100.0%        |
|     VP fixing rate      |        100.0%        |
+-------------------------+----------------------+


+------------------------+--------------------+-------------------+------------------
+----------------------+
| Refine Place Movements | Mean displacement | Max displacement | Instances Moved
|       ERROR ID       |
+------------------------+--------------------+-------------------+------------------
+----------------------+
|     Certus_Client0     |      0.00 um      |      0.00 um      |       N/A
|         N/A          |
|     Certus_Client1     |        N/A        |        N/A        |       N/A
|         N/A          |
|     Certus_Client2     |      0.00 um      |      0.00 um      |       N/A
|         N/A          |
+------------------------+--------------------+-------------------+------------------
+----------------------+

Failure Reasons for Hold Opt:
+---------------------------------------------------+
|               No Violations found                 |
+---------------------------------------------------+

#######################################################Final Routing
DRC######################################################################
+----------------+-----------------------+
|Certus Client(s)|  Final DRCvio call#1  |
+----------------+-----------------------+
| certus client0 |          0            |
| certus client1 |          0            |
| certus client2 |          0            |
+----------------+-----------------------+


#####################################################Change in
Density###########################################################################
+---------------+-----------------+-----------------+
|Certus Client(s)| Initial Density |  Final Density  |
+---------------+-----------------+-----------------+
|certus client0 |     1.365%      |     2.444%      |
```

```
|certus client1 |     0.812%      |     0.812%      |
|certus client2 |     18.075%     |     18.686%     |
+---------------+-----------------+-----------------+

###################################################DIAGs & ERROR
ID###########################################################################
+------------------+-----------------------------------------------------------
---------------------------------------------------------------------------------
-------+
|     Process      |
          DIAGs
      |
+------------------+-----------------------------------------------------------
---------------------------------------------------------------------------------
-------+
|  Certus manager  | N/A

      |
|  certus_client0  | N/A

      |
|  certus_client1  | N/A

      |
|  certus_client2  | N/A

      |
|  tempus_client0  | N/A

      |
|  tempus_client1  | N/A

      |
+------------------+-----------------------------------------------------------
---------------------------------------------------------------------------------
-------+
+------------------+----------------+
|     Process      |    ERROR ID    |
+------------------+----------------+
|  Certus manager  | IMPEXT-7082    |
|  certus_client0  | IMPESO-301     |
|  certus_client1  | IMPESO-301     |
```

```
|  certus_client2  | IMPESO-301     |
|  tempus_client0  | IMPEXT-7082    |
|  tempus_client1  | IMPEXT-7082    |
+-----------------+----------------+
```

- The following command generates the summary report, `certus.logv1.summary.txt`, for the specified log file, `certus.logv1`:

  ```
  certus> generate_summary -logvfilename certus.logv1
  *info: Generating summary report from log:  ['/workspace/data/certus.logv1']
  *info: summary report generated in certus.logv1.summary.txt

  Certus summary report: certus.logv1.summary.txt
  ```

- The following command generates the output file, `out.txt`, which includes the log summary details for the specified log file, `certus.logv1`.

  ```
  certus> generate_summary -logvfilename certus.logv1 -outputfile out.txt
  *info: Generating summary report from log:  ['/workspace/data/certus.logv1']
  *info: summary report generated in out.txt

  Certus summary report: out.txt
  ```

Related Commands

- `opt_signoff`
- `start_distributed_eco_clients`
- `time_design_signoff`

get_db

```
get_db
[-help]
[design_flow_effort]
[design_power_effort]
```

Returns the current settings of the `set_db` command.

Parameters

`-help`	Outputs a brief description that includes type and default information for each `get_db` parameter.
`parameter_names`	Displays information for the specified parameters. See the `set_db` command for descriptions of the `set_db` parameters that you can specify.

Examples

- The following command displays the current settings of `set_db design_flow_effort`:

  ```
  certus > get_db design_flow_effort
  design_flow_effort standard # enums={express standard extreme}, default=standard
  standard
  ```

- The following command displays the current settings of `set_db design_power_effort`:

  ```
  certus > get_db design_power_effort
  design_power_effort none # enums={none low high}, default=none
  none
  ```

Related Commands

- `set_db`

implement_distributed_eco

```
implement_distributed_eco
[-help]
[-continue]
[-force]
```

Performs ECO routing, parasitic extraction, and parasitic back-annotation on the Certus client manager as well as the Tempus clients, followed by a timing update. When you run this command, the Certus client manager session is ready for any signoff timing analysis post ECO implementation, and you can start the next session of ECO closure.

Parameters

`-help`	Outputs a brief description that includes type and default information for each `implement_distributed_eco` parameter.
`-continue`	Allows the `implement_distributed_eco` command to continue where the previous call of that command failed. This can be useful in scenarios where parasitic extraction failed due to some missing data. Using this parameter, you can then update the current session and restart the `implement_distributed_eco` command to perform only parasitic extraction.
`-force`	Performs ECO routing and parasitic extraction on all the blocks (except the blocks that are not loaded on a Certus client). By default, the `implement_distributed_eco` command performs ECO routing and parasitic extraction only on the blocks where ECO changes have been made.

Examples

- The following command runs incremental routing and parasitic extraction on each Certus client that was modified by ECO optimization step, and back-annotates the parasitic on the Tempus timing sessions:

certus> implement_distributed_eco

- In case `implement_distributed_eco` errors out during parasitic extraction, you can update a few settings on the Certus clients and restart from the `extractRC` step using the following commands:

```
certus> set_db opt_signoff_distributed_optimization_pre_extraction_tcl
pre_extractRC.tcl
```

```
certus> implement_distributed_eco -continue
```

Related Commands

- `start_distributed_eco_clients`

opt_signoff

```
opt_signoff
[-help]
[-all]
[-area]
[-drv]
[-dynamic]
[-hold]
[-leakage]
[-power]
[-report_dir]
[-report_full_clock_path]
[-setup]
[-update_timing]
```

Let's you choose between the timing closure flow or the timing and power closure flow in Certus. In addition, it honors the signoff timing from Tempus clients and performs the physical-aware distributed optimization through each Certus client.

Note: This command requires a 64-bit executable.

Parameters

`-help`	Outputs a brief description that includes type and default information for each `opt_signoff` parameter. For a detailed description of the command and all its parameters, use the `man` command: `man opt_signoff`.
`-all`	Performs DRV, Setup, Hold, and Power optimizations in the Path-based Analysis (PBA) mode. Also, this parameter enables ECO routing and parasitic extraction using the `implement_distributed_eco` command, and provides signoff timing analysis using the `time_design_signoff` command. **Note**: • Aggressive Setup optimization at the

	expense of Hold is supported if you use the `set_db design_flow_effort extreme` setting. • Performs power closure along with timing closure if `set_db design_power_report low/high` setting is used. • Performs aggressive power optimization if you use the `set_db design_power_report high` setting. • Performs only timing closure (DRV/Setup/Hold) as a default flow. • Prints a detailed timing report using the `time_design_signoff` command, which includes the full clock path when the `-report_full_clock_path` parameter is specified along with `opt_signoff -all`. You can specify the directory to save these reports using the `-report_dir` parameter. • Prints a detailed power summary report using the `time_design_signoff -report_power` command. • This parameter does not replace all customized fine-tuned flows. It provides the Certus users with easy access to Timing and Power ECO closure technology. However, the existing Certus flow using individual optimization steps within Certus is still supported.
`-area`	Reclaims area on the netlist without causing any DRV/Setup/Hold degradation.

`-drv`	Optimizes DRV violations. DRV optimization fixes violations in the following order: • `glitch` • `drv (max_transition/max_capacitance violations)` • `xtalk delta delay` **Note**: • DRV optimization supports buffer addition and resizing. • CellEM violations are fixed only when `set_db opt_signoff_fix_cell_em` is set to `true`.
`-dynamic`	Prints the initial and final power numbers. It optimizes the dynamic power to reduce the total power consumption.
`-hold`	Optimizes the hold violations.

`-leakage`	Optimizes leakage power. Leakage optimization supports only VT swapping.
	Note: To optimize leakage using resizing in addition to VT swapping, use the following:
	`set_db opt_signoff_power_opt_focus`
	`opt_signoff -power`

`-power`	Optimizes total power in the design. During total power optimization, the software performs swapping and sizing on both the combinational and sequential cells, and also does buffer removal when enabled. It reduces leakage, internal, and switching power in the design.
`-report_dir string`	Specifies the directory to save reports.

`-report_full_clock_path`	Generates a detailed timing report that contains the full clock path.
`-setup`	Optimizes the setup timing violations. The setup optimization supports buffer addition, VT swapping, and resizing.
`-update_timing`	Generates updated timing data for use during optimization.

Examples

- The following commands perform timing closure (DRV, Setup, Hold) in PBA mode:

  ```
  certus> set_db design_power_effort none design_flow_effort standard
  certus> opt_signoff -all
  ```

- The following commands perform timing and power closure (DRV, Setup, Hold, and Power) in PBA mode:

  ```
  certus> set_db design_power_effort low/high design_flow_effort standard
  certus> opt_signoff -all
  ```

- The following commands perform timing and power closure in PBA mode, and generate the full clock path report in the directory specified by the user with `-report_dir`:

  ```
  certus> set_db design_power_effort low/high design_flow_effort standard
  certus> opt_signoff -all -report_full_clock_path -report_dir <>
  ```

- The following commands perform timing closure with aggressive Setup fixing at the expense of Hold timing in PBA mode:

  ```
  certus> set_db design_power_effort none design_flow_effort extreme
  certus> opt_signoff -all
  ```

- The following command performs Hold fixing while not degrading the Setup timing:

  ```
  certus> opt_signoff -hold
  ```

- The following command recovers area by performing cell downsizing and buffer deletion while ensuring not to create any new Setup/Hold/DRV violations.

  ```
  certus> opt_signoff -area
  ```

Related Commands

- `implement_distributed_eco`
- set_db
- `time_design_signoff`

opt_signoff Category Attributes

The root attributes in the `opt_signoff` category are:

- opt_signoff_custom_distribution_file
- opt_signoff_distributed_optimization_block_restore_file
- opt_signoff_distributed_optimization_max_clients_per_host
- opt_signoff_distributed_optimization_out_dir
- opt_signoff_distributed_optimization_pre_eco_route_tcl
- opt_signoff_distributed_optimization_post_eco_route_tcl
- opt_signoff_distributed_optimization_pre_extraction_tcl
- opt_signoff_distributed_optimization_post_extraction_tcl
- opt_signoff_distributed_optimization_post_restore_design_tcl
- opt_signoff_distributed_optimization_pre_restore_design_tcl

opt_signoff_custom_distribution_file

```
opt_signoff_custom_distribution_file filename
```

Default: ""

Specifies a file that provides customized resource distribution information for CPU and memory configurations of Certus and Tempus clients. This attribute supports various resource requirements, helping ensure optimal performance.

Specify the following custom resource distribution details in `opt_signoff_custom_distribution_file:`

- For every client, either for Tempus or Certus, or Tempus and Certus, specify the following:
  `module=<moduleName> view=<viewName> nCpu=<Number of threads allowed> {<farm distribution instruction>}`
- For client machine sharing, specify the View as `view=<viewName>` with one (or multiple) Modules, `module=<moduleName>`
- For two Certus clients sharing the same machine, specify `module=<moduleName>` twice on the same line (where the `nCPU` value is then divided by the number of Certus clients and provided

to each Certus client).

Note:

This attribute must be specified before `distribute_read_db`.

Example:

The following command refers to a file (`distribution.tcl`) that refers to each client requesting a separate machine:

```
opt_signoff_custom_distribution_file distribution.tcl
```

where (`distribution.tcl`) file includes:

```
module=topChip nCpu=16 {bsub -q batch -W 100:00 -P SSV:26.1:Certus -o /dev/null -n 8 -R
'select\[OSREL==EE80\] rusage\[mem=250000,tmp=50000\] span\[hosts=1\]'}
module=small_cpu nCpu=8 {bsub -q batch -W 100:00 -P SSV:26.1:Certus -o /dev/null -n 8 -
R 'select\[OSREL==EE80\] rusage\[mem=150000,tmp=50000\] span\[hosts=1\]'}
module=big_GPU nCpu=16 {bsub -q batch -W 100:00 -P SSV:26.1:Certus -o /dev/null -n 16 -
R 'select\[OSREL==EE80\] rusage\[mem=250000,tmp=50000\] span\[hosts=1\]'}
view=setup_view1 nCpu=16 {bsub -q batch -W 100:00 -P SSV:26.1:Certus -o /dev/null -n 16
-R 'select\[OSREL==EE80\] rusage\[mem=200000,tmp=100000\] span\[hosts=1\]'}
view=setup_view2 nCpu=16  {bsub -q batch -W 100:00 -P SSV:26.1:Certus -o /dev/null -n
16 -R 'select\[OSREL==EE80\] rusage\[mem=200000,tmp=100000\] span\[hosts=1\]'}
view=setup_hold1 nCpu=8  {bsub -q batch -W 100:00 -P SSV:26.1:Certus -o /dev/null -n 8
-R 'select\[OSREL==EE80\] rusage\[mem=200000,tmp=100000\] span\[hosts=1\]'}
view=setup_power  nCpu=8  {bsub -q batch -W 100:00 -P SSV:26.1:Certus -o /dev/null -n 8
-R 'select\[OSREL==EE80\] rusage\[mem=200000,tmp=100000\] span\[hosts=1\]'}
```

For details, see the Recommended Guidelines for Certus chapter in the *Certus Stylus User Guide.*

opt_signoff_distributed_optimization_block_restore_file

```
opt_signoff_distributed_optimization_block_restore_file  filename
```

Provides a list of pointers to the Innovus database that can be loaded on Certus clients when calling the `start_distributed_eco_clients` command. If some partitions in the Certus client manager netlist are not part of the block restore file, those partitions will be considered don’t touch during ECO closure. If the Certus client manager has a reduced database when running in High Capacity ECO mode, partitions not present in the Certus client manager will not be loaded by any Certus client.

Example:

```
blockName1 INVS_DB_PATH_for_block1
```

`blockName2 INVS_DB_PATH_for_block2`

Any partition that is listed as part of `opt_signoff_dont_touch_partition_list` or `opt_signoff_touch_partition_list` or `opt_signoff_intra_partition` or `opt_signoff_inter_partition` or `opt_signoff_inter_partition_exclusive` must be specified in the block restore file; however, for a few partitions that are excluded from ECO, only the partition name needs to be specified.

Example:

`blockName3` is a nested partition inside `blockName2`. If Signoff ECO optimization is not allowed inside `blockName3`, and `opt_signoff_intra_partition` partition fixing is applied to `blockName2` only, then you must specify the partition name, `blockName3`, in the block restore file, and skip specifying its Innovus database:

```
blockName1 INVS_DB_PATH_for_block1

blockName2 INVS_DB_PATH_for_block2

blockName3
```

opt_signoff_distributed_optimization_max_clients_per_host

`opt_signoff_distributed_optimization_max_clients_per_host value`

Default: 0

When a value greater than 0 is provided, the Certus client manager is instructed to reuse the Tempus client's machines when starting Certus clients using the `start_distributed_eco_clients` command. This allows sharing the CPUs between Tempus clients and Certus clients.

Note that memory usage on those machines is equal to the sum of one of the Tempus client process memory and N Certus client process memory, where N is the value set by this parameter.

opt_signoff_distributed_optimization_out_dir

`opt_signoff_distributed_optimization_out_dir  directoryname`

Specifies the directory name where the Certus clients' log files will be generated. By default, the tool uses `DistOpt/` in the current directory.

opt_signoff_distributed_optimization_pre_eco_route_tcl

`opt_signoff_distributed_optimization_pre_eco_route_tcl  filename`

Provides a TCL script that each Certus client will source before performing ECO routing during the `implement_distributed_eco` command.

opt_signoff_distributed_optimization_post_eco_route_tcl

```
opt_signoff_distributed_optimization_post_eco_route_tcl  filename
```

Provides a TCL script that each Certus client will source after performing ECO routing during the `implement_distributed_eco` command.

opt_signoff_distributed_optimization_pre_extraction_tcl

```
opt_signoff_distributed_optimization_pre_extraction_tcl  filename
```

Provides a TCL script that each Certus client will source before performing parasitic extraction during the `implement_distributed_eco` command.

opt_signoff_distributed_optimization_post_extraction_tcl

```
opt_signoff_distributed_optimization_post_extraction_tcl  filename
```

Provides a TCL script that each Certus client will source after performing parasitic extraction during the `implement_distributed_eco` command.

opt_signoff_distributed_optimization_post_restore_design_tcl

```
opt_signoff_distributed_optimization_post_restore_design_tcl  filename
```

Provides a TCL script that each Certus client will source after loading the Innovus DB.

opt_signoff_distributed_optimization_pre_restore_design_tcl

```
opt_signoff_distributed_optimization_pre_restore_design_tcl filename
```

Provides a TCL script that each Certus client will source before loading the Innovus DB.

Note: The above `opt_signoff` attributes are specific to Certus only. For details related to other attributes of the `opt_signoff` command, refer to the *Tempus Stylus Text Reference*

Manual document.

Examples

The following command refers to a file (`blocks.txt`) pointing to all Innovus DBs that are opened for ECO:

```
set_db opt_signoff_distributed_optimization_block_restore_file blocks.txt
```

Below is a template for the `blocks.txt` file:

```
blockName1 <dirName>/blockName1.db

blockName2 <dirName>/blockName2.db
```

For sample scripts, refer to the *Template Scripts* section of the *Certus Stylus User Guide*.

Related Commands

- `start_distributed_eco_clients`

set_db

```
set_db
[-help]
[reset_db -category design; reset_db -category add_fillers]
[design_flow_effort]
[design_power_effort]
```

This command, when used in combination with `opt_signoff -all`, enables you to choose between the timing closure flow or the timing and power closure flow in Certus cockpit, where `opt_signoff -all` honors the signoff timing from the Tempus clients and performs the physical-aware distributed optimization through each Certus client.

Parameters

`-help`	Outputs a brief description that includes type and default information for each `set_db` parameter.

`design_flow_effort {express|standard|extreme}`	Guides the main command's effort (`opt_signoff -all`) for timing closure. • `express`: In the Certus cockpit, this flow runs similarly to the standard flow. • `standard`: Configures the flow to improve the quality of results and the full-flow turnaround time. This flow is useful for most designs. • `extreme`: Configures the flow to produce higher accuracy results, reducing the turnaround time (TAT) significantly. This flow is useful for designs where timing closure is challenging. *Default*: `standard`
`design_power_effort {none|low|high}`	Specifies the effort to reduce the dynamic and leakage power. The tradeoff between dynamic and leakage power can also be controlled. • `none`: No power optimization is performed, and power is not considered during timing optimization. • `low`: Power is considered during timing optimization, and power-aware optimization is performed in all flow stages. • `high`: Additional algorithms are used to minimize the power consumption. The turnaround time with this option will be longer than using `low` or `none`. *Default*: `none`

`reset_db`	Resets the attribute value to its predefined default value. It returns a single value when resetting a single attribute, a list of values when resetting the same attribute on different object instances, or a human-readable table for multiple attributes in the following format: `Attribute_name   value`

Examples

- The following command performs Timing closure (DRV, SETUP, HOLD) in PBA mode:

```
certus> set_db design_power_effort none design_flow_effort standard
certus> opt_signoff -all
```

- The following command performs Timing and Power Closure (DRV, SETUP, HOLD, and POWER) in PBA mode:

```
certus> set_db design_power_effort low/high design_flow_effort standard
certus> opt_signoff -all
```

- The following command performs Timing Closure with aggressive Setup fixing at the expense of Hold timing in PBA mode:

```
certus> set_db design_power_effort none design_flow_effort extreme
certus> opt_signoff -all
```

Related Commands

- `opt_signoff`
- `get_db`

start_distributed_eco_clients

```
start_distributed_eco_clients
[-help]
```

Launches the Certus clients, which will each load the respective Innovus database. The tool depends on the input file specified by `opt_signoff_distributed_optimization_block_restore_file` to obtain pointers to these Innovus databases. If some partitions in the Certus client manager netlist are not part of the block restore file, they are considered don't-touch during ECO closure.

When the Certus client manager has a reduced database while running in High Capacity ECO mode, partitions not present in the Certus client manager will not be loaded by any Certus client. Certus clients can either collect their own machines or reuse Tempus clients' machines when specifying `opt_signoff_distributed_optimization_max_clients_per_host` with a value greater than 0. The Certus clients are instructed by `set_multi_cpu_usage -cpu_per_remote_host` to identify how many threads can be used by the clients. In case the Certus clients have to collect more machines from the farm, you have to specify the correct farm requests using the `set_distributed_host` command.

Parameters

`-help`	Prints out the command usage.

Related Commands

- `implement_distributed_eco`

time_design_signoff

```
time_design_signoff
[-help]
[-no_expanded_views]
[-report_dir string]
[-report_prefix string]
[-report_full_clock_path]
[-report_power]
```

Generates signoff timing reports for each view from the Tempus clients. The command prints out a timing summary table per view and all the views combined.

Parameters

`-help`	Outputs a brief description that includes type and default information for each `time_design_signoff` parameter.
`-no_expanded_views`	Disables printing of expanded timing per view.
`-report_dir` *string*	Specifies the directory to save reports.
`-report_prefix` *string*	Specifies the prefix for timing report file names.
`-report_full_clock_path`	When this parameter is specified, the detailed timing report generated includes the full clock path.
`-report_power`	When this parameter is specified, `time_design_signoff` prints a detailed power summary report from the power view.

Examples

- The following command prints the Setup/Hold/DRV timing summary:

```
certus> time_design_signoff
```

```
--------------------------------------------------------------------------------
                      time_design_signoff Summary
--------------------------------------------------------------------------------

+--------------------+-----------+-----------+-----------+-----------+-----------+
|     Setup mode     |   all     | reg2reg   |  in2reg   | reg2out   |  in2out   |
+--------------------+-----------+-----------+-----------+-----------+-----------+
|           WNS (ns):|    -1.797 |    -0.536 |    -1.797 |    -0.244 |       N/A |
|           TNS (ns):| -2054.234 | -266.725 | -1798.030 |    -0.244 |       N/A |
|    Violating Paths:|      8735 |      3402 |      5402 |         1 |         0 |
+--------------------+-----------+-----------+-----------+-----------+-----------+
|funct-rcMax         |    -1.797 |    -0.536 |    -1.797 |    -0.244 |       N/A |
|                    | -2054.234 | -266.725 | -1798.030 |    -0.244 |       N/A |
|                    |      8735 |      3402 |      5402 |         1 |         0 |
+--------------------+-----------+-----------+-----------+-----------+-----------+

+----------------+------------------------------+--------------------------------+
|                |         Signal nets          |           Clock nets           |
|   DRVs         +------------------+-----------+------------------+-------------+
|                |  Nr nets(terms)  | Worst Vio |  Nr nets(terms)  | Worst Vio   |
+----------------+------------------+-----------+------------------+-------------+
|   max_cap      |     16 (     16)|     -0.346 |      0 (      0) |          0  |
|   max_tran     |    442 (    690)|     -2.117 |      0 (      0) |          0  |
|   max_fanout   |      0 (      0)|          0 |      0 (      0) |          0  |
+----------------+------------------+-----------+------------------+-------------+

+--------------------+-----------+-----------+-----------+-----------+-----------+
|      Hold mode     |   all     | reg2reg   |  in2reg   | reg2out   |  in2out   |
+--------------------+-----------+-----------+-----------+-----------+-----------+
|          WNS (ns): |    -0.362 |    -0.362 |     0.143 |    -0.294 |       N/A |
|           TNS (ns):|  -340.889 |  -256.428 |     0.000 |   -84.461 |       N/A |
|   Violating Paths: |      1773 |      1262 |         0 |       511 |         0 |
+--------------------+-----------+-----------+-----------+-----------+-----------+
|funct-rcMin         |    -0.362 |    -0.362 |     0.143 |    -0.294 |       N/A |
|                    |  -340.889 |  -256.428 |     0.000 |   -84.461 |       N/A |
|                    |      1773 |      1262 |         0 |       511 |         0 |
+--------------------+-----------+-----------+-----------+-----------+-----------+
```

- The following command reports power summary from the power view:

```
certus> time_design_signoff -report_power
```

```
--------------------------------------------------------------------------------
                      time_design_signoff Summary
--------------------------------------------------------------------------------
```

```
+--------------------+-----------+-----------+-----------+-----------+-----------+
|     Setup mode     |   all     | reg2reg   |  in2reg   | reg2out   |  in2out   |
+--------------------+-----------+-----------+-----------+-----------+-----------+
|           WNS (ns):|    -1.797 |    -0.536 |    -1.797 |    -0.244 |       N/A |
|           TNS (ns):| -2054.234 |  -266.725 | -1798.030 |    -0.244 |       N/A |
|    Violating Paths:|      8735 |      3402 |      5402 |         1 |         0 |
+--------------------+-----------+-----------+-----------+-----------+-----------+
|funct-rcMax         |    -1.797 |    -0.536 |    -1.797 |    -0.244 |       N/A |
|                    | -2054.234 |  -266.725 | -1798.030 |    -0.244 |       N/A |
|                    |      8735 |      3402 |      5402 |         1 |         0 |
+--------------------+-----------+-----------+-----------+-----------+-----------+

+----------------+-------------------------------+------------------------------+
|                |         Signal nets           |           Clock nets         |
|   DRVs         +-----------------+-------------+-----------------+------------+
|                |  Nr nets(terms) | Worst Vio   |  Nr nets(terms)  | Worst Vio  |
+----------------+-----------------+-------------+-----------------+------------+
|   max_cap      |     16 (     16)|     -0.346 |      0 (      0) |         0  |
|   max_tran     |    442 (    690)|     -2.117 |      0 (      0) |         0  |
|   max_fanout   |      0 (      0)|          0 |      0 (      0) |         0  |
+----------------+-----------------+-------------+-----------------+------------+

+--------------------+-----------+-----------+-----------+-----------+-----------+
|      Hold mode     |   all     | reg2reg   |  in2reg   | reg2out   |  in2out   |
+--------------------+-----------+-----------+-----------+-----------+-----------+
|          WNS (ns): |    -0.362 |    -0.362 |     0.143 |    -0.294 |       N/A |
|           TNS (ns):|  -340.889 |  -256.428 |     0.000 |   -84.461 |       N/A |
|   Violating Paths: |      1773 |      1262 |         0 |       511 |         0 |
+--------------------+-----------+-----------+-----------+-----------+-----------+
|funct-rcMin         |    -0.362 |    -0.362 |     0.143 |    -0.294 |       N/A |
|                    |  -340.889 |  -256.428 |     0.000 |   -84.461 |       N/A |
|                    |      1773 |      1262 |         0 |       511 |         0 |
+--------------------+-----------+-----------+-----------+-----------+-----------+

Power report summary using power view :  funct-rcMax
Total Power
-------------------------------------------------------------------------------------
-----
Total Internal Power:     1130.32884617           62.2254%
Total Switching Power:     667.26277622           36.7333%
Total Leakage Power:        18.91418979            1.0412%
Total Power:              1816.50581218
-------------------------------------------------------------------------------------
-----
```

```
Group                              Internal   Switching     Leakage       Total
Percentage
                                   Power      Power         Power         Power
(%)
------------------------------------------------------------------------------------
-----
Sequential                            675.1       95.71       4.636       775.5
42.69
Macro                                 123.4       6.658        3.46       133.5
7.349
IO                                        0           0           0
0           0
Combinational                           284       491.3       10.39       785.7
43.26
Clock (Combinational)                 24.94       67.43      0.2621       92.63
5.099
Clock (Sequential)                    22.83       6.155      0.1692       29.16
1.605
------------------------------------------------------------------------------------
-----
Total                                  1130       667.3       18.91        1817
   100
------------------------------------------------------------------------------------
-----
```

Related Commands

- export_distributed_eco

2

Global Configuration Variables

eco_certus_client_install_dir

`eco_certus_client_install_dir`

Default: " "

Specifies an Innovus version for the Certus client. The Certus client installation directory specified by this TCL variable overwrites any other path specified for the Certus client using any other mechanism, including $PATH.

You need to provide the path to the `<install_dir>/lnx86` directory of the Certus client (Innovus) installation.

To set this global variable, use the `set_db` command.

Note: This global variable must be specified before `opt_signoff_distributed_optimization_block_restore_file`.

Examples

- The following command allows you to point to the Certus client installation directory:

  ```
  certus> set_db eco_certus_client_install_dir
  /icd/flow/INNOVUS/INNOVUS231/latest.isr1/lnx86
  ```

- The following command provides a list of pointers to the Innovus database that can be loaded on Certus clients:

  ```
  certus> set_db opt_signoff_distributed_optimization_block_restore_file
  blkRestore.txt
  ```

- The following command restores all design views from the `saveDir` directory, and generates view-specific reports in directories with the corresponding view name within the `pdfDir` directory:

  ```
  certus> distribute_read_db -read_db all -read_dir saveDir -out_dir pdfDir
  ```

3

Manual ECO Commands and Attributes

replay_distributed_eco

```
replay_distributed_eco
[-help]
-block_eco_file_list <string>
```

Supports the Non-Interactive Manual ECO mode. This command sources the Innovus-compliant ECO files that are specified as input using `block_eco_file_list <file>`. The command specified in these ECO files will first be executed on the respective Certus clients before being processed for execution on the manager. After this, the processed commands will be executed (block by block) on both the manager and the Tempus clients.

Parameters

`-help`	Outputs a brief description that includes type and default information for each `replay_distributed_eco` parameter. For a detailed description of the command and all its parameters, use the man command: `man replay_distributed_eco`.

`-block_eco_file_list <string>`	Specifies the list of Certus client block module names and the corresponding ECO files as input. You can also specify multiple ECO files for the same module.
	
	Note:
	
	You must provide the input files in the sequence of the optimization calls. The format of the file is presented below:
	
	`<block-module-1> <path-to-block-1-ECO-file>`
	
	`<block-module-2> <path-to-block-2-ECO-file>`
	
	…
	
	`<block-module-N> <path-to-block-N-ECO-file>`

Examples

- The following command refers to a file (`blockecofile`) pointing to all Innovus-compliant Manual ECO TCL files that are to be replayed :

```
certus> replay_distributed_eco -block_eco_file_list blockecofile
```

Below is a template for `blockecofile`:

```
blockName1 blockName1_innovus.tcl
blockName2 blockName2_innovus.tcl
```

Refer to the Sample Template Script section of the *Certus Stylus User Guide*.

4

System Commands - Basic Tool-Tcl Commands

gui_show

```
gui_show
[-help]
[-designs <string>]
```

Displays the Certus Manager GUI.

The `gui_show -designs` command launches the GUI of the Certus Clients.

Parameters

`-help`	Prints a brief description that includes type and default information for each `gui_show` parameter. For a detailed description of the command and all of its parameters, use the `man` command: `man gui_show`

`-designs <string>`	Specifies the list of partitions for which `gui_show` is executed.
	Note:
	The `gui_show -designs` command can be run at any point after `start_distributed_eco_clients` is specified. The tool waits for the design to be loaded on the Certus Client(s) before launching the GUI. The Certus Manager shell remains inaccessible until all Certus Clients GUI are closed. This GUI should be used as a viewer only to view the physical related constraints such as placement blockage, routing blockage, routing congestion, local density, and the availability of legal locations.
	The following warning is displayed when you run `gui_show -designs` after `start_distributed_eco_clients`:
	`Operations done through the Certus Clients GUI should not impact routing, should not change dont_touch/dont_use attribute, should not trigger a new Parasitic extraction or a reset of parasitic, should not perform any timing report or timing query.`
	`Purpose of this GUI is to view the layout, query any physical attribute/property from the Certus Clients or change some instances placement status, and anything else is prohibited as it can desynchronize the Certus client manager from the Certus clients.`

Examples

- The following command launches the GUI for the Certus Clients for the specified partitions, `ptn_wrapper` and `tdsp_core:`

  ```
  certus> gui_show -designs {ptn_wrapper tdsp_core}
  ```

- The following command launches the GUI for all the partitions loaded as Certus clients:

  ```
  certus> gui_show -designs all
  ```

Related Commands

- `start_distributed_eco_clients`